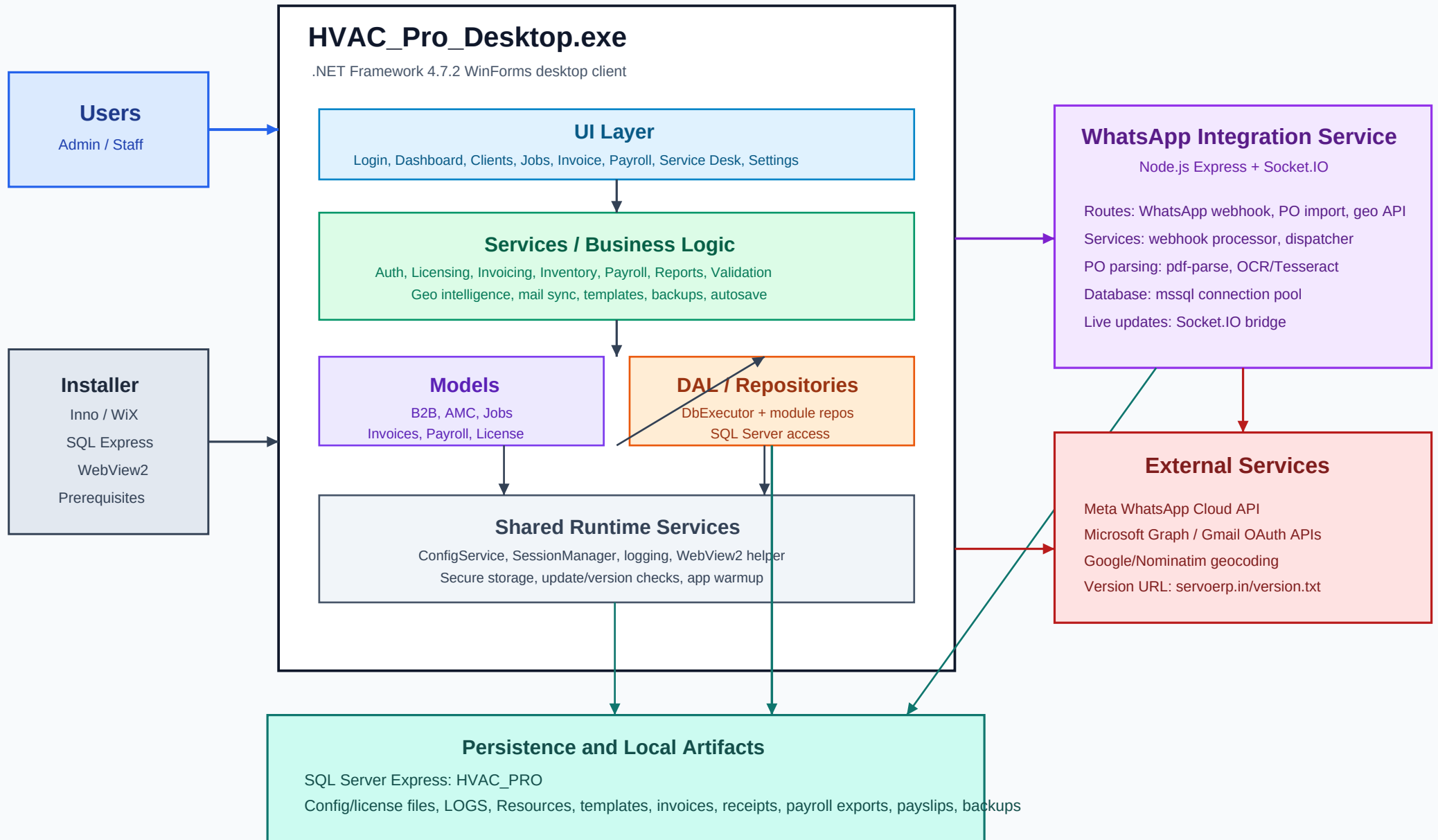


HVAC Pro / ServoERP Application Architecture

Derived from SOURCE_CODE/HVAC_Pro_Desktop.csproj, HVACPro.config, and whatsapp-integration-service/package.json



Legend: arrows show primary runtime calls/data flow; boxes group source folders and deployment artifacts.

Architecture Notes

This page summarizes the major responsibilities visible in the repository.

Desktop client

- SOURCE_CODE builds HVAC_Pro_Desktop.exe as a .NET Framework 4.7.2 WinForms application.
- UI forms and user controls host the main workflows: dashboard, clients, contracts, invoices, payments, jobs, payroll, inventory, service desk, geo intelligence, reports, settings, login, and licensing.

Business and platform services

- SERVICES contains module services plus cross-cutting services for auth, session management, licensing, audit trail, validation, templates, document rendering, backups, autosave, logging, update checks, mail sync, and WebView2 runtime checks.
- MODELS defines the shared domain objects for B2B clients, contacts, sites, AMC contracts, invoices, stock, employees, jobs, payroll, service desk, mail integration, master data, and licensing.

Data access

- DAL uses DatabaseManager, DbExecutor, and module repositories to read and write SQL Server data.
- The configured default database is HVAC_PRO on local SQLEXPRESS, with tenant and app settings in HVACPro.config.

Local outputs

- The app writes operational artifacts around the workspace/application folder: LOGS, Invoice, RECEIPTS, PAYSLIPS, PAYROLL_EXPORTS, Resources, CONFIG, and license/config files.

Integrations

- MailIntegrationService connects to Outlook via Microsoft Graph and Gmail via Google APIs using OAuth, then turns incoming messages into service desk updates or incidents.
- Geo services call Google geocoding and Nominatim, with a WebView2 embedded map resource under Resources/GeoMap/map.html.
- The app checks servoerp.in/version.txt for update/version information when enabled.

WhatsApp integration service

- whatsapp-integration-service is a Node.js service using Express, Socket.IO, mssql, multer, pdf-parse, pg, and Tesseract.js.
- It exposes health, WhatsApp webhook, purchase-order import, geo intelligence, and send-template routes; it connects to SQL Server and Meta WhatsApp Cloud API and emits live updates over Socket.IO.

Deployment

Generated file: Docs/HVAC_Pro_App_Architecture.pdf

- Installer scripts and WiX/Inno setup files package the desktop app, SQL Server Express prerequisites, WebView2 fixes, update packages, and enterprise installer/VM